Quantification: 876_z89c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	9
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2175.11	2169	869.225	px^2
major_axis_length_px	lacuna shape	84.7631	81.6741	19.9666	px
minor_axis_length_px	lacuna shape	33.7022	35.8518	7.7268	px
aspect_ratio	lacuna shape	2.557	2.7987	0.4604	unitless
eccentricity	lacuna shape	0.9107	0.934	0.04	unitless
solidity	lacuna shape	0.8171	0.8169	0.0544	unitless
roots_count	headline	12.6667	12	5.244	count
ring_length_r30_px	headline	731.111	724	110.188	px
ring_length_r60_px	reported	1938.56	2098	364.245	px
owned_length_px	ownership-dependent	6565.7	6817.45	2401.13	px
edge_count	ownership-dependent	488.667	498	179.776	count
mean_edge_length_px	ownership-dependent	13.4523	13.467	0.3903	px
perimeter_px	normalised	294.887	274.676	88.6524	px
ring_area_r30_px2	normalised	9333.67	9180	1991.24	px^2
ring_area_r60_px2	normalised	23018.1	23485	4492.97	px^2
in_frame_fraction_r30	normalised	0.9828	1	0.0515	unitless
in_frame_fraction_r60	normalised	0.9539	1	0.0857	unitless
ring_density_r30	normalised	0.0798493	0.0796067	0.0123574	px^-1
ring_density_r60	normalised	0.0846482	0.0821726	0.0103971	px^-1
roots_per_100px_perimeter	normalised	4.2577	4.2819	1.0987	per 100 px
ring_attached_length_r30_px	network v2	658.556	700	161.254	px
ring_attached_length_r60_px	network v2	1751.67	1904	491.98	px
ring_length_w_r30_px	network v2	823.7	824.671	125.287	px
ring_length_w_r60_px	network v2	2196.23	2373	417.583	px
sholl_crossings_r10	network v2	16.2222	17	3.5978	count
sholl_crossings_r20	network v2	19.8889	20	4.285	count
sholl_crossings_r30	network v2	22.7778	24	4.8933	count
loop_count	loops, report only	80.6667	81	36.6742	count
loop_edge_share	loops, report only	0.5629	0.5663	0.096	share
ring_width_mean_r30_px	width	5.9702	5.6859	0.7533	px

Per field
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measure	value	units
analysed_area_px2	1.01997e+06	px^2
total_skeleton_length_px	77625	px
canalicular_length_density_per_px	0.0761049	px^-1
junction_count	2869	count
junction_density_per_px2	0.00281282	px^-2
lacuna_count	11	count

lacunae_per_px2	1.078e-05	px^-2
skeleton_component_count	544	count
mean_component_length_px	142.693	px
median_component_length_px	10	px
field_length_density_w_per_px	0.085756	px^-1
field_density_without_flagged_per_px	0.0765645	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	90	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0761049	px^-1
field_length_density_with_unattached_per_px	0.0761931	px^-1
loop_count	964	count
loop_edge_share	0.5343	share
edge_count	6088	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2967	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	62933	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2536	138.359	30.0256	4.608	0.9762	0.657	-1.519	12.5193	411.862	True
2	6490	249.288	41.8559	5.9559	0.9858	0.5841	-1.5551	22.8282	886.791	True
3	3313	115.919	39.5634	2.93	0.94	0.7245	-1.5598	154.062	372.366	False
4	2754	105.554	35.8518	2.9442	0.9406	0.7558	1.5255	165.057	102.925	False
5	3301	102.585	44.3824	2.3114	0.9016	0.8169	-1.4051	307.062	909.902	False
6	2169	78.4025	38.343	2.0448	0.8723	0.8413	1.557	311.008	321.631	False
7	956	61.2609	21.1134	2.9015	0.9387	0.8	1.5273	461.978	61.1475	False
8	2610	87.8807	38.6915	2.2713	0.8979	0.8782	-1.531	535.001	198.764	False
9	1500	59.3512	33.3189	1.7813	0.8276	0.8966	1.5421	545.937	440.713	False
10	1670	81.6741	26.9572	3.0298	0.944	0.8337	-1.4297	692.65	612.224	False
11	1303	70.2403	25.0978	2.7987	0.934	0.8068	-1.3129	914.738	39.4635	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	17	698	1406	2392.42	174	13.7495	
2	34	1057	2018	7840.01	569	13.7786	
3	23	870	2298	6817.45	498	13.6897	5.2883
4	16	829	2103	5050.13	375	13.467	5.6859
5	16	705	2099	9589.3	699	13.7186	5.4671
6	12	807	2098	5321.55	399	13.3372	6.7128
7	12	724	1641	3740.62	290	12.8987	6.9903
8	13	753	2188	7507.19	575	13.056	6.7179

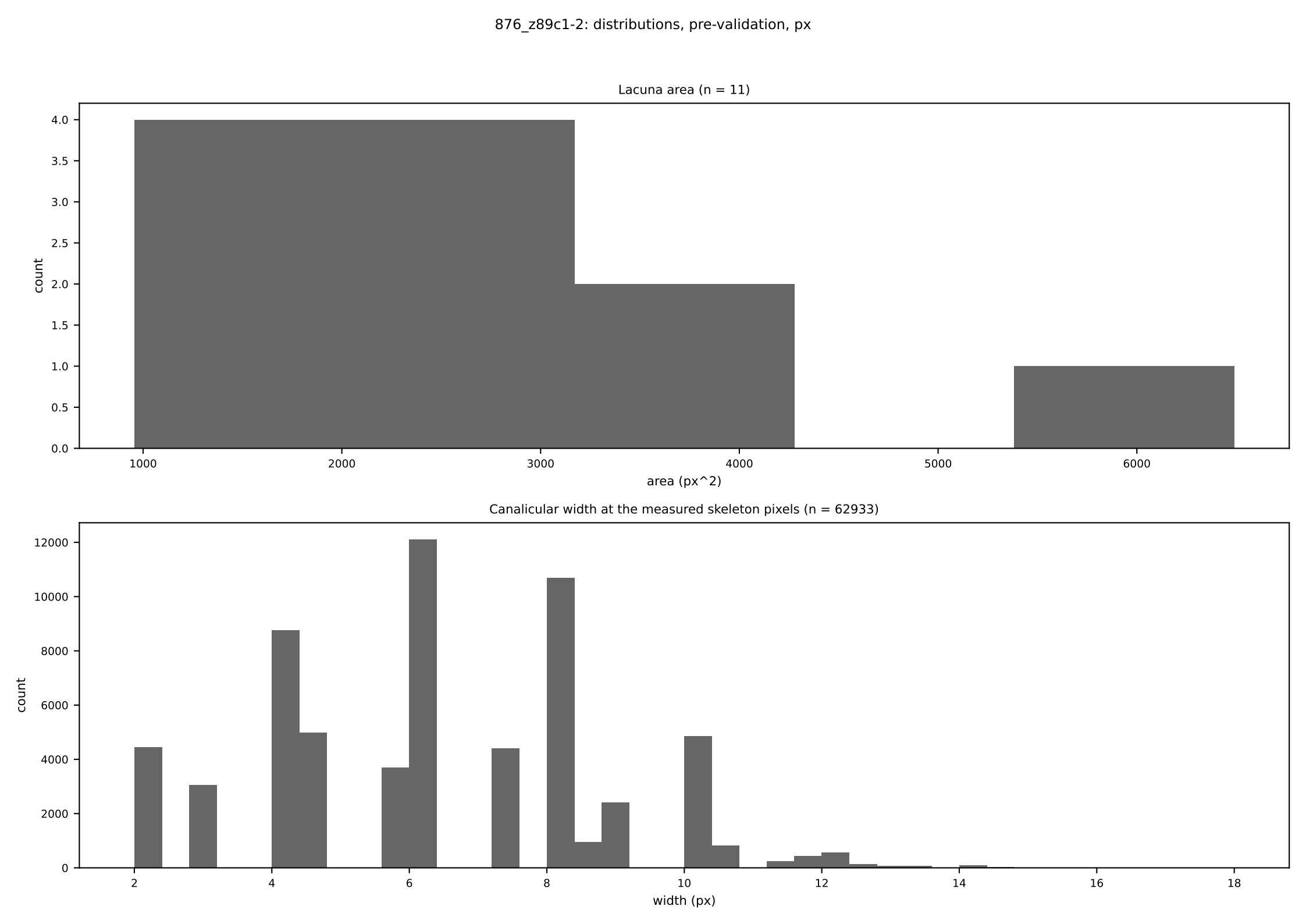
9	6	708	2031	9832.18	753	13.0573	6.3834
10	9	697	1882	8038.6	581	13.8358	5.6684
11	7	487	1107	3194.32	228	14.0102	4.818

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	583.204	7886	16742	0.6082	0.5704	0.0885113	0.0839804	2.9149
2	1053.68	14114	25792	0.6408	0.5406	0.0748902	0.0782413	3.2268
3	423.014	12381	28912	1	1	0.070269	0.0794826	5.4372
4	413.328	11320	26755	1	0.9628	0.0732332	0.0786021	3.871
5	373.665	11365	27711	1	0.9875	0.0620326	0.0757461	4.2819
6	273.505	9180	23731	1	1	0.0879085	0.0884076	4.3875
7	191.315	7221	16849	1	0.887	0.100263	0.0973945	6.2724
8	277.848	9459	23485	1	1	0.0796067	0.0931658	4.6788
9	190.593	7507	20585	1	1	0.094312	0.0986641	3.1481
10	274.676	8714	22903	1	1	0.0799862	0.0821726	3.2766
11	236.043	6856	16232	0.8455	0.748	0.0710327	0.0681986	2.9656

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	662	1282	804.097	1599.04	19	17	16
2	1037	1910	1178.44	2232.76	29	19	22
3	797	2215	966.482	2572.16	19	22	28
4	758	1904	929.625	2380.85	22	23	24
5	532	1693	785.37	2373	14	22	26
6	781	1922	915.683	2378.7	18	20	22
7	685	1550	824.671	1845	18	18	22
8	731	2182	861.754	2505.51	15	20	26
9	700	2008	806.399	2323.93	17	17	22
10	659	1711	783.671	2148.56	13	26	24
11	284	580	539.647	1238.39	10	11	11


876_z89c1-2: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 62933)
count
width (px)